

Vector Equations

Geometric Algorithms
Lecture 4

Practice Problem

Suppose that A is a 322×245 augmented matrix for a system with infinitely many solutions. What is the maximum number of pivot positions that A can have?

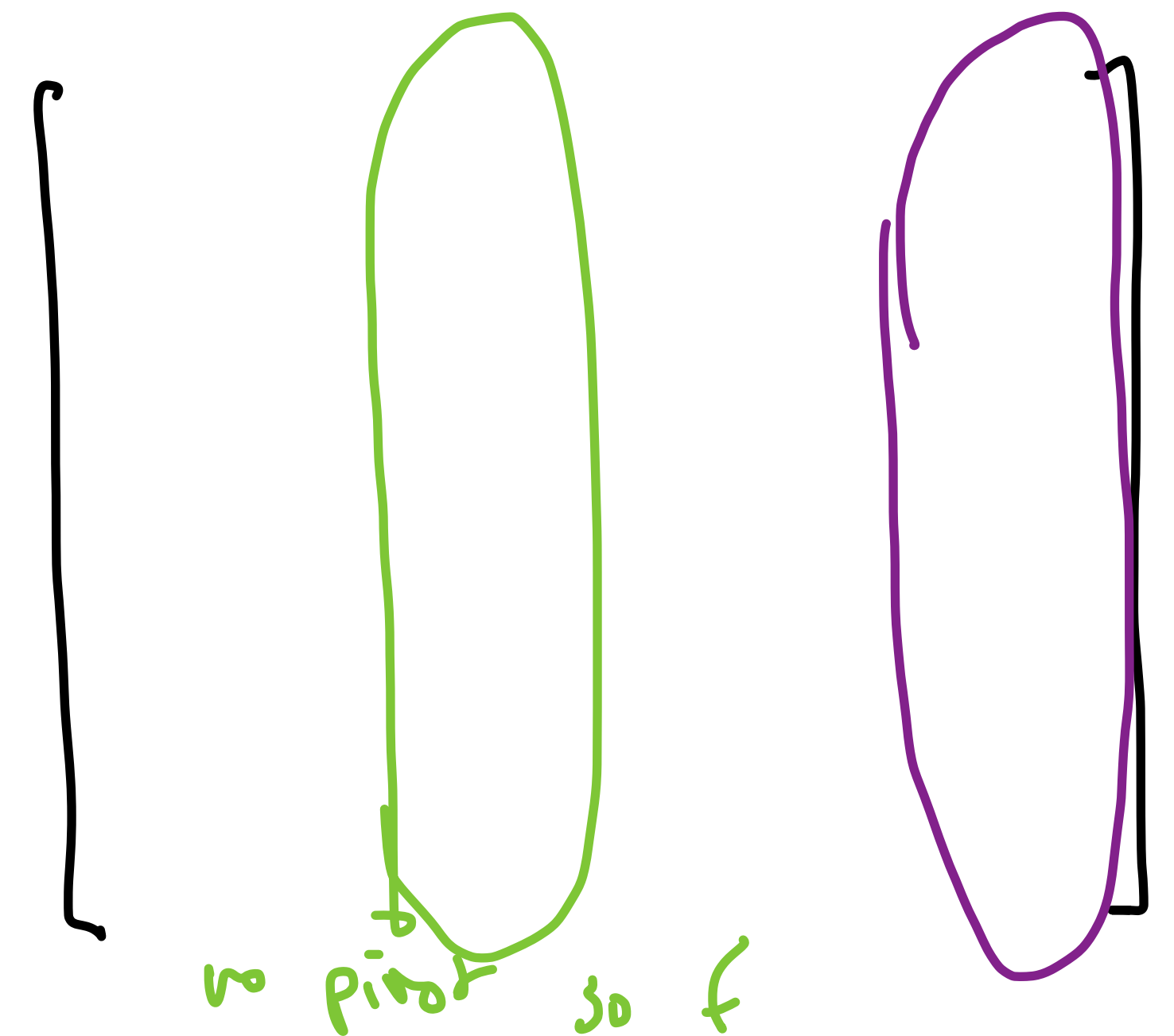
243

244

321

245

$A \sim$



Answer

$\underbrace{6}_{\text{rows}} \times \underbrace{4}_{\text{columns}}$ matrix

$$\begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \end{bmatrix} \sim \begin{bmatrix} 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$0x_1 + 0x_2 + 0x_3 = 1 \quad 0 = 1$$

A is aug. mat. of inf. sol. system

Answer

$A \sim$

aug. mat. of inf. sol. system

$$\left[\begin{array}{ccc|c} \boxed{1} & 0 & 0 & a \\ 0 & \boxed{\times} & 0 & b \\ 0 & 0 & \boxed{1} & c \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

free

$$\begin{aligned} x_1 &= a \\ x_2 &= b \\ x_3 &= c \end{aligned}$$

Answer

2×5
~~2~~

$$\begin{bmatrix} \boxed{1} & + & \boxed{2} & \boxed{2} & + \\ + & \boxed{2} & \boxed{2} & \boxed{2} & + \end{bmatrix}$$

free

A is $(m \times n)$ matrix s.t. A is aug. mat.

rows cols

of inf. sol. system.

max prob pos: $\min(m, n-2)$

Outline

- » Formally define vectors
- » Discuss vector operations and vector algebra
- » Draw the connection between vectors and systems of linear equations

Keywords

vector

vector addition

vector scaling/multiplication

the zero vector

vector equations

linear combinations

span

A Bit of Motivation

A Bit of Motivation

Data are ordered collections of values

A Bit of Motivation

Data are ordered collections of values

Q: When are two data *similar*?

A Bit of Motivation

Data are ordered collections of values

Q: When are two data *similar*?

A: When they're **close together** geometrically

A Note on Algebra

$$\mathbf{v} = \mathbf{w}$$

$$\mathbf{v} + \mathbf{w}$$

$$a\mathbf{v}$$

A Note on Algebra

In programming, an *interface* is an abstract collection of related functions

$$\mathbf{v} = \mathbf{w}$$

$$\mathbf{v} + \mathbf{w}$$

$$a\mathbf{v}$$

A Note on Algebra

In programming, an *interface* is an abstract collection of related functions

$$\mathbf{v} = \mathbf{w}$$

And object then *implements* an interface

$$\mathbf{v} + \mathbf{w}$$

$$a\mathbf{v}$$

A Note on Algebra

In programming, an *interface* is an abstract collection of related functions

$$\mathbf{v} = \mathbf{w}$$

And object then *implements* an interface

$$\mathbf{v} + \mathbf{w}$$

$$a\mathbf{v}$$

Doing abstract algebra is like implementing an *algebraic* interface

Vectors

What is a vector (in $\mathbb{R}^n$)?

A. an n -tuple of real numbers

B. a point in $\mathbb{R}^n$

C. a 1-column matrix with real values

D. all of the above

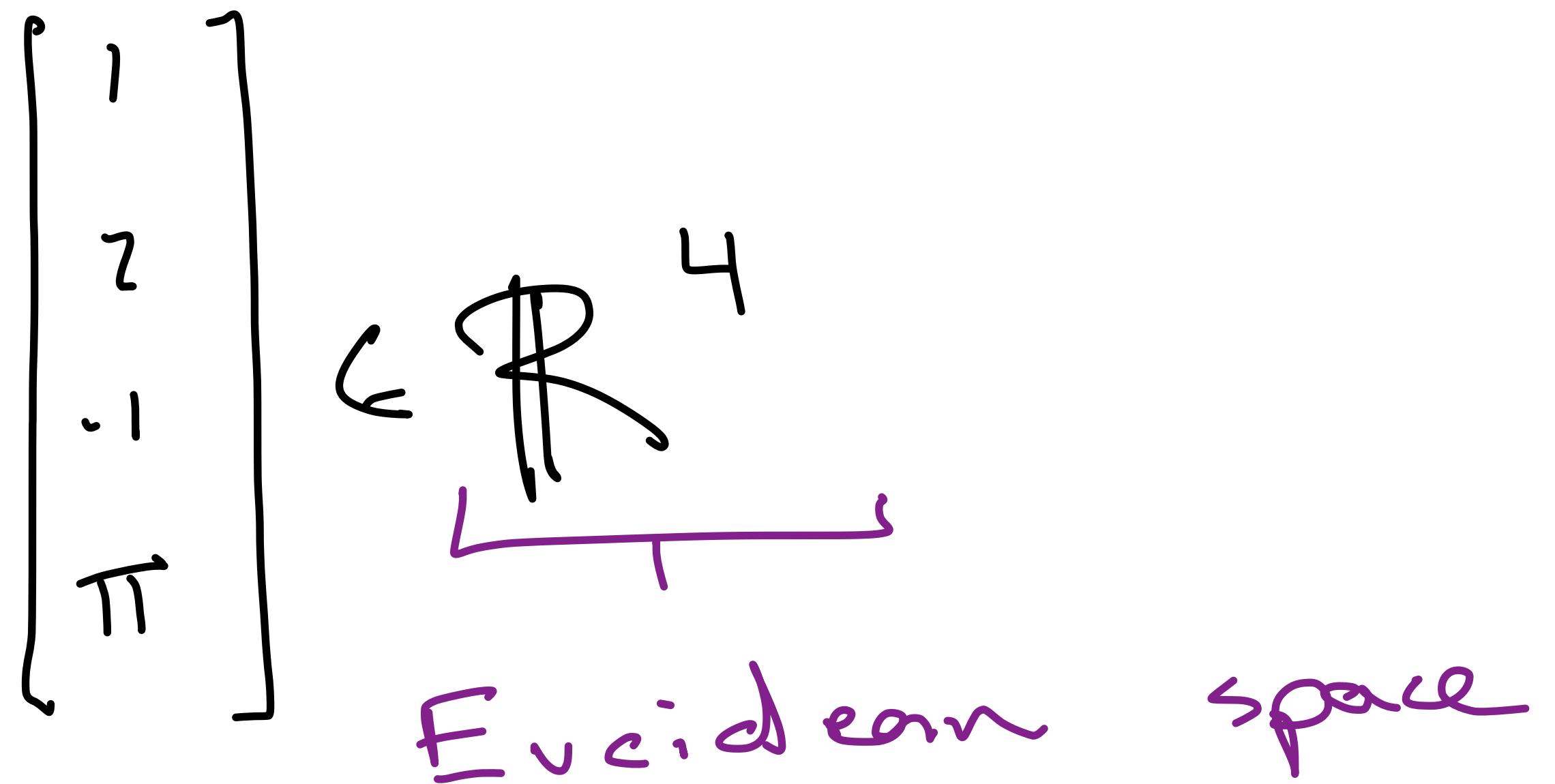
E. none of the above?

Column Vectors

Df. a *column vector* is a matrix with a single column

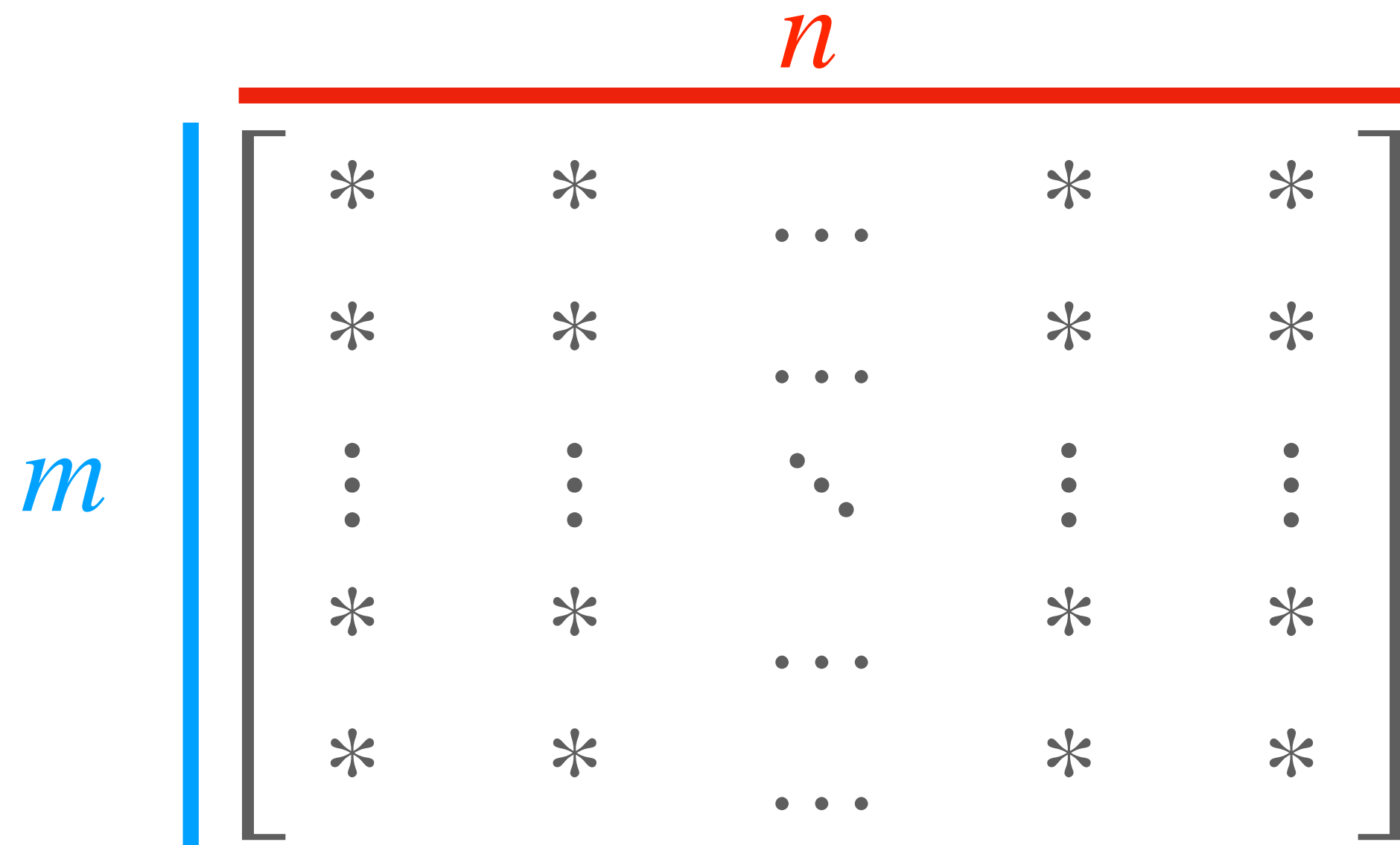
$$\begin{bmatrix} 1 \\ 2 \\ -1 \\ \pi \end{bmatrix} \in \mathbb{R}^4$$

Euclidean space

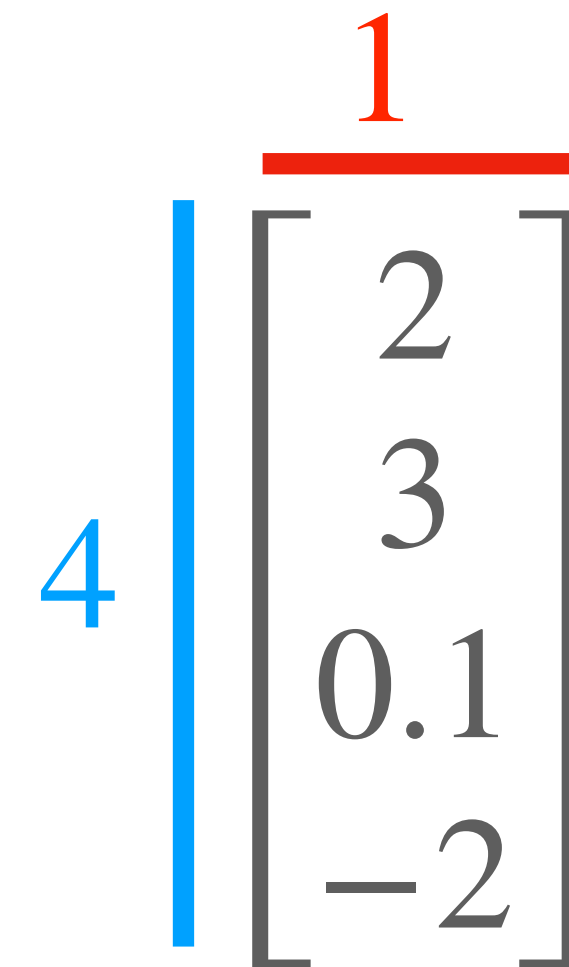
A hand-drawn diagram illustrating a column vector in a 4D Euclidean space. On the left, a column vector is represented as a matrix with four rows and one column, containing the values 1, 2, -1, and π. To the right of the vector, there is a handwritten symbol for a 4D real vector space, R^4, with a double arrow pointing from the vector to it. Below this, a purple bracket is drawn, and the words "Euclidean space" are written in purple cursive script.

Matrix Size

An $(m \times n)$ matrix has m rows and n columns



A diagram of a general matrix with m rows and n columns. A blue vertical line to the left of the matrix is labeled with the variable m in blue. A red horizontal line above the matrix is labeled with the variable n in red. The matrix is enclosed in large square brackets and contains asterisks (*) and ellipses (...) to represent its elements.



A diagram of a column vector with 4 rows and 1 column. A blue vertical line to the left of the vector is labeled with the number 4 in blue. A red horizontal line above the vector is labeled with the number 1 in red. The vector is enclosed in large square brackets and contains the numerical values 2, 3, 0.1, and -2.

Matrix Size

An $(m \times n)$ matrix has m rows and n columns

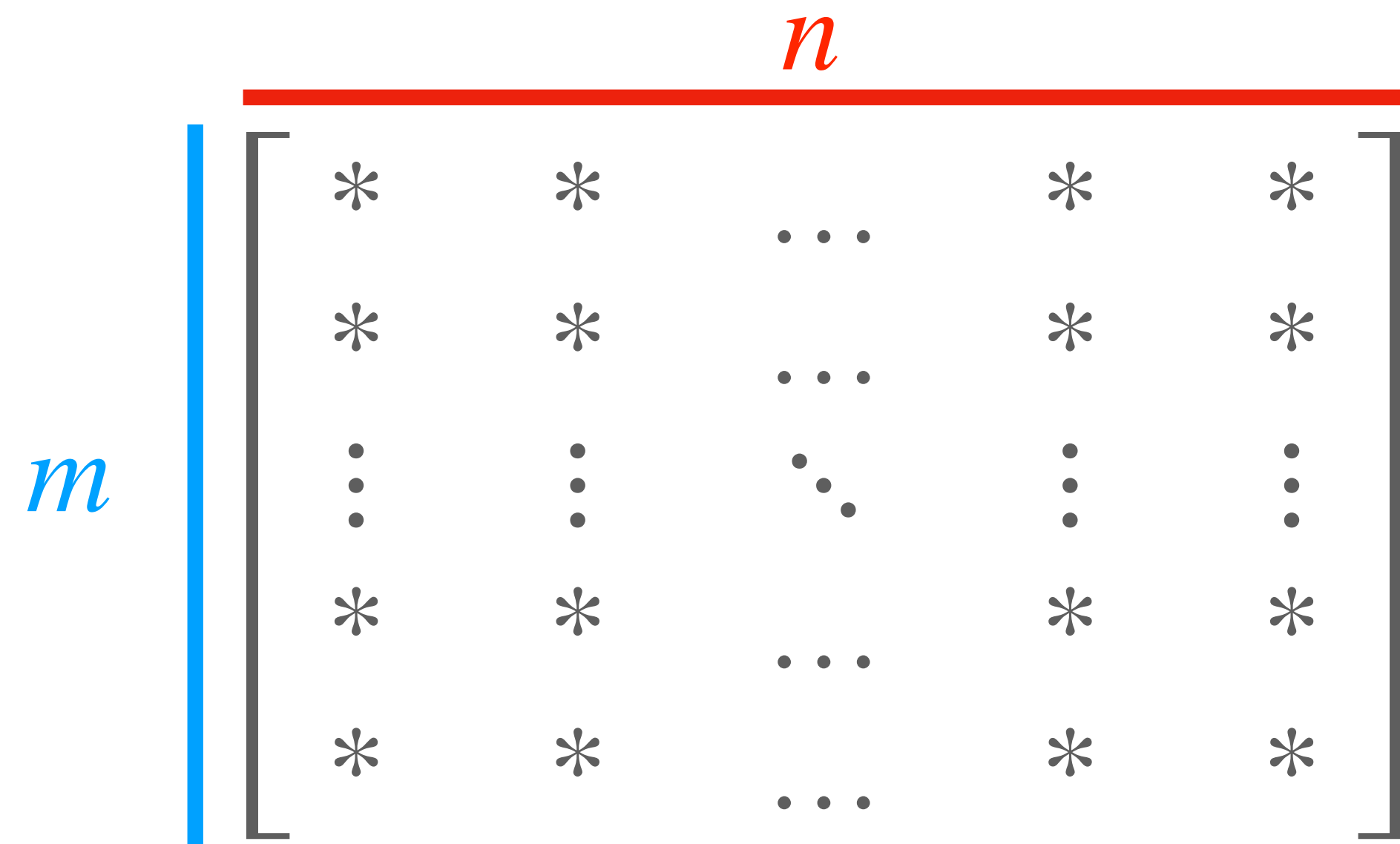
$$\begin{matrix} & \overbrace{\hspace{10em}}^n \\ \underbrace{\hspace{1em}}_m \left[\begin{array}{ccccc} * & * & \dots & * & * \\ * & * & \dots & * & * \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ * & * & \dots & * & * \\ * & * & \dots & * & * \end{array} \right] \end{matrix}$$

$$\begin{matrix} & \overbrace{\hspace{1em}}^1 \\ \underbrace{\hspace{1em}}_4 \left[\begin{array}{c} 2 \\ 3 \\ 0.1 \\ -2 \end{array} \right] \end{matrix} \hookrightarrow \mathbb{R}^4 \quad \cancel{\mathbb{R}^{4 \times 1}}$$

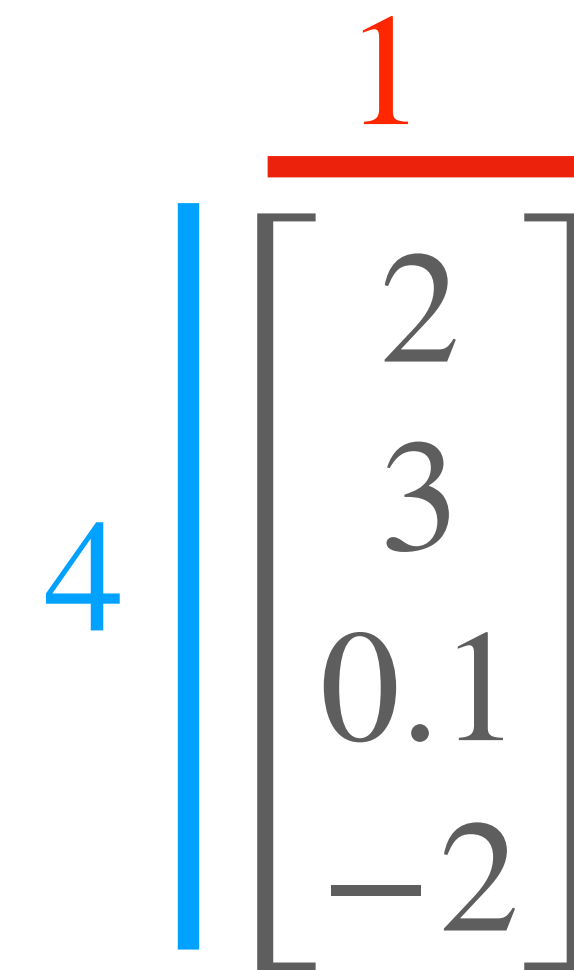
$\mathbb{R}^{m \times n}$ is set of matrices with $\mathbb{R}$ entries

Matrix Size

An $(m \times n)$ matrix has m rows and n columns



A diagram of a general matrix with m rows and n columns. A blue vertical line to the left of the matrix is labeled with a blue italic m . A red horizontal line above the matrix is labeled with a red italic n . The matrix is enclosed in large square brackets and contains asterisks (*) and ellipses (...) representing its entries.

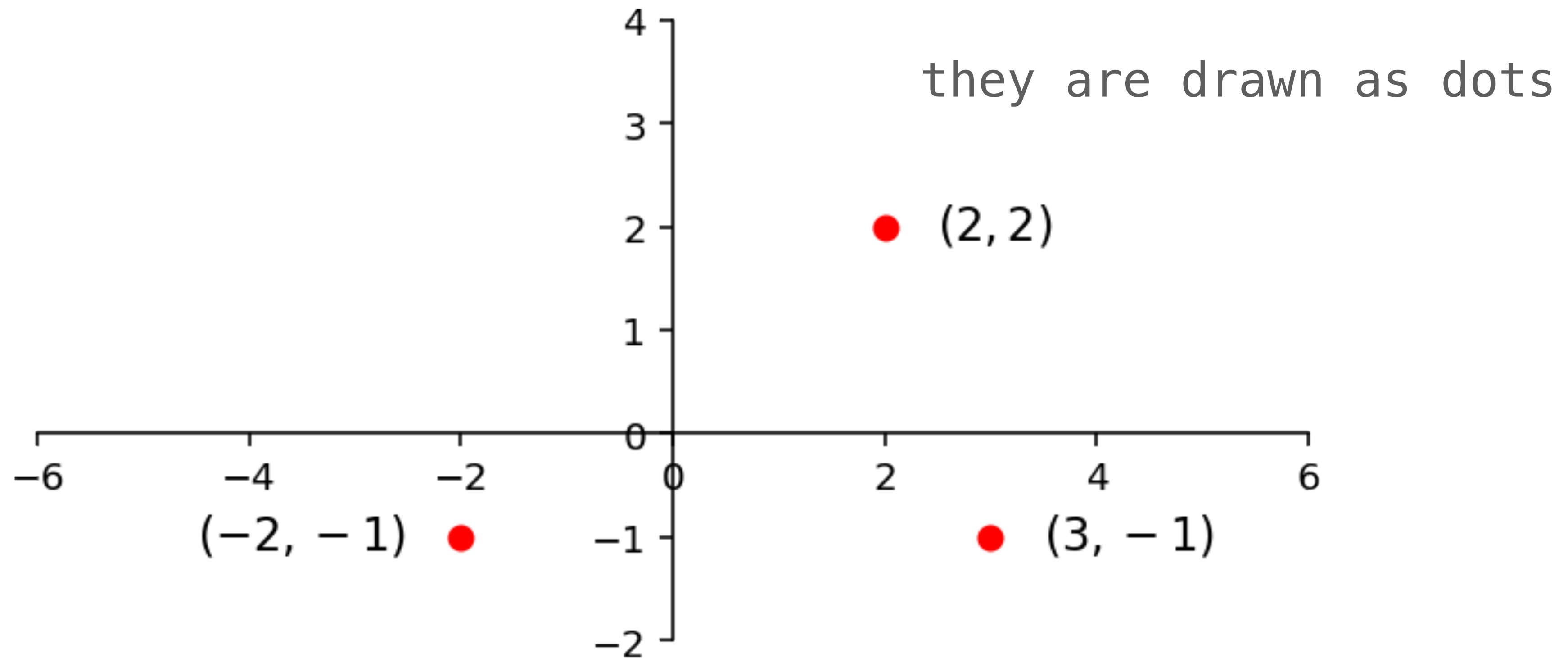


A diagram of a vector with 4 rows and 1 column. A blue vertical line to the left of the vector is labeled with a blue number 4. A red horizontal line above the vector is labeled with a red number 1. The vector is enclosed in large square brackets and contains the numerical values 2, 3, 0.1, and -2.

the number of rows
of a vectors is
called its **dimension**

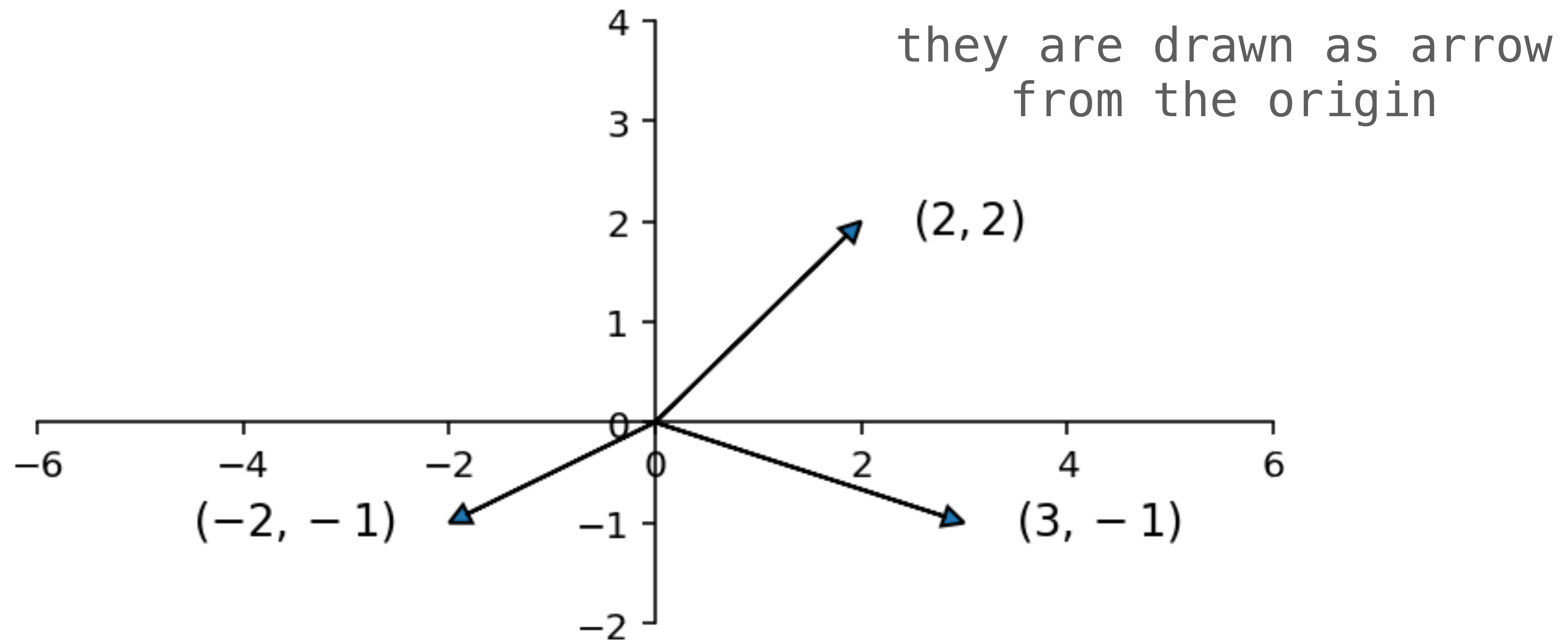
$\mathbb{R}^{m \times n}$ is set of matrices with $\mathbb{R}$ entries

Notation (Points)



points in $\mathbb{R}^2$ are notated as (a, b)

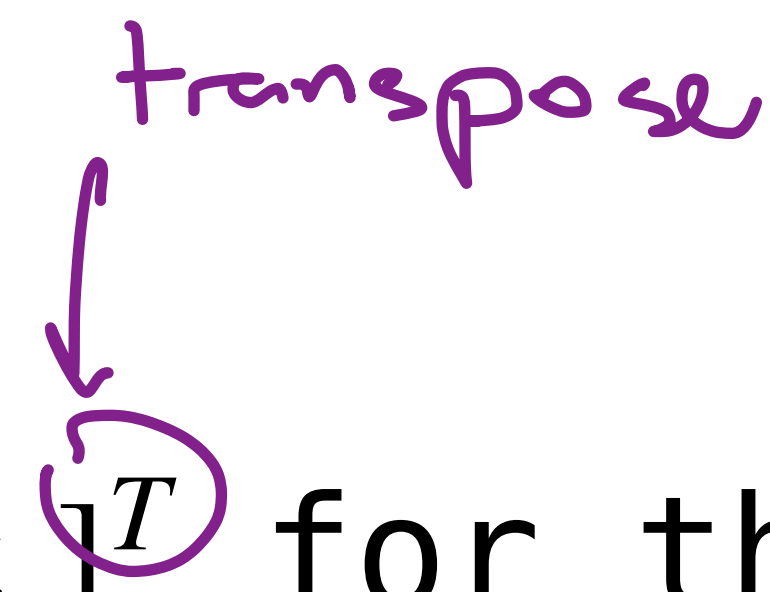
Notation (Vectors)



vectors in $\mathbb{R}^2$ are notated as $\begin{bmatrix} a \\ b \end{bmatrix}$

Notation (Looking ahead)

We will often write $[a_1 \ a_2 \ \dots \ a_n]^T$ for the vector $\begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{bmatrix}$



Notation (Looking ahead)

We will often write $[a_1 \ a_2 \ \dots \ a_n]^T$ for the vector $\begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{bmatrix}$

IMPORTANT

$(a_1, a_2, \dots, a_n)$ is not the same as $\underset{\text{row}}{[a_1 \ a_2 \ \dots \ a_n]}$

$\in \mathbb{R}^{1 \times n} \neq \mathbb{R}^n$

$\underset{\text{vector}}$

Vector Operations

Vector "Interface"

Vector "Interface"

equality what does it mean for two vectors
to be equal?

Vector "Interface"

equality what does it mean for two vectors
to be equal?

addition what does $\mathbf{u} + \mathbf{v}$ (adding two vectors
mean?

Vector "Interface"

equality what does it mean for two vectors to be equal?

addition what does $\mathbf{u} + \mathbf{v}$ (adding two vectors mean?

scaling what does $a\mathbf{v}$ (multiplying a vector by a real number) mean?

Vector "Interface"

equality what does it mean for two vectors to be equal?

addition what does $\mathbf{u} + \mathbf{v}$ (adding two vectors mean?

scaling what does $a\mathbf{v}$ (multiplying a vector by a real number) mean?

What properties do they need to satisfy?

Vector "Interface"

equality what does it mean for two vectors to be equal?

addition what does $\mathbf{u} + \mathbf{v}$ (adding two vectors mean?

scaling what does $a\mathbf{v}$ (multiplying a vector by a real number) mean?

What properties do they need to satisfy?

Vector Equality

$$\begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

$$\begin{bmatrix} 1 \\ 2 \end{bmatrix} \neq \begin{bmatrix} 3 \\ 4 \end{bmatrix}$$

~~$$\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 4 \\ 5 \end{bmatrix}$$~~

must be
the same

$$\begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix}$$

if and only if
($\Leftrightarrow$)
(iff)

$$x_1 = y_1$$

and

$$x_2 = y_2$$

and

$\vdots$

$$x_n = y_n$$

Vector Equality

two vectors are equal if their entries at each position are equal

(this is also the case for matrices)

!!IMPORTANT!!
ORDER MATTERS

$$\begin{bmatrix} 1 \\ 2 \end{bmatrix} \neq \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

Vector Equality

$$\begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix} \quad \text{is the same as} \quad \begin{array}{l} a_1 = b_1 \\ a_2 = b_2 \\ \vdots \\ a_n = b_n \end{array}$$

Vector "Interface"

equality what does it mean for two vectors to be equal?

addition what does $\mathbf{u} + \mathbf{v}$ (adding two vectors mean?

scaling what does $a\mathbf{v}$ (multiplying a vector by a real number) mean?

What properties do they need to satisfy?

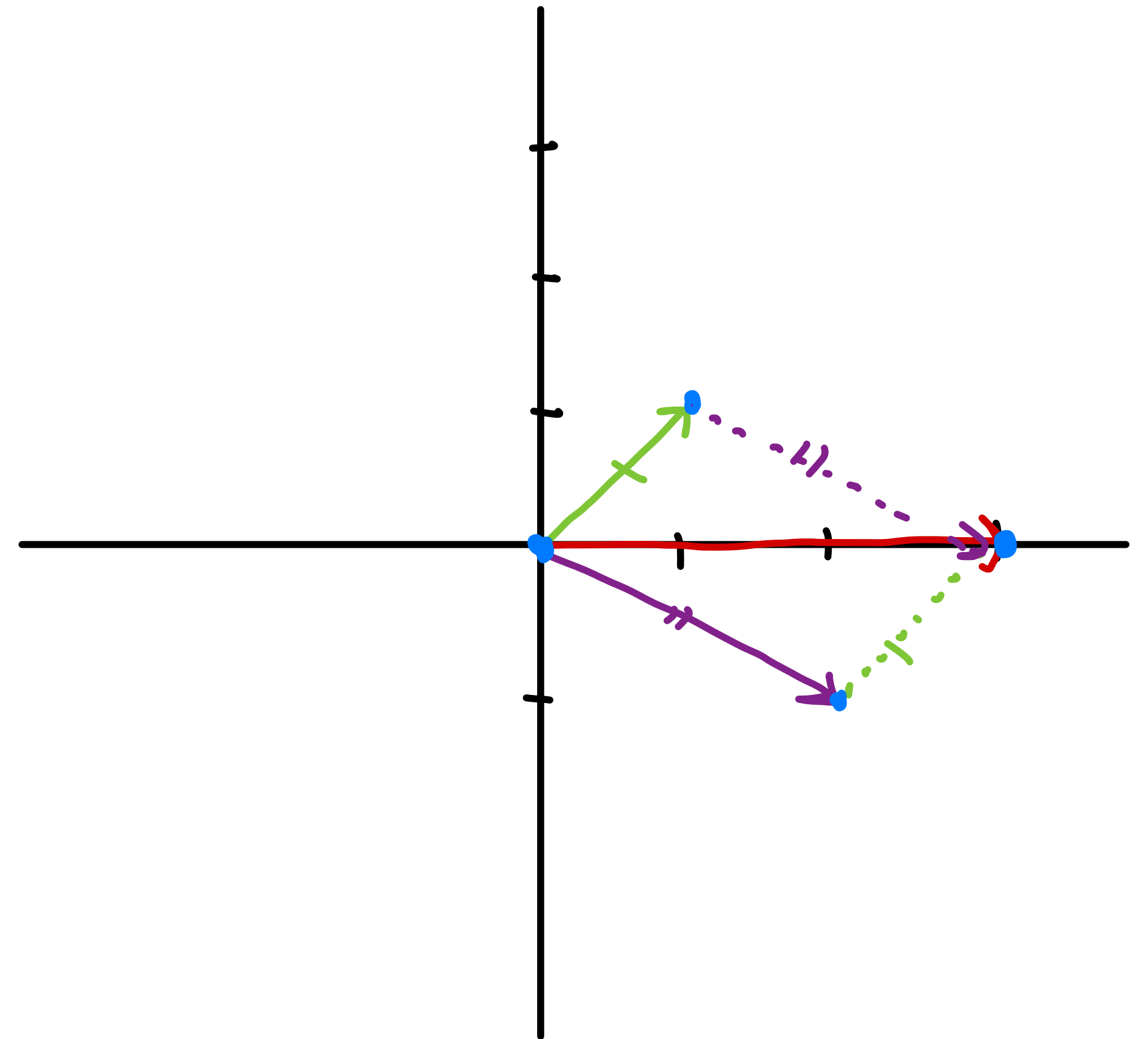
Vector Addition

$$\begin{bmatrix} 1 \\ 2 \end{bmatrix} + \begin{bmatrix} 3 \\ 4 \end{bmatrix} = \begin{bmatrix} 1+3 \\ 2+4 \end{bmatrix} = \begin{bmatrix} 4 \\ 6 \end{bmatrix}$$

$$\begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} + \begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix} = \begin{bmatrix} x_1 + y_1 \\ \vdots \\ x_n + y_n \end{bmatrix}$$

must be same

$$\begin{bmatrix} 1 \\ 1 \end{bmatrix} + \begin{bmatrix} 2 \\ -1 \end{bmatrix} = \begin{bmatrix} 3 \\ 0 \end{bmatrix}$$



Vector Addition

adding two vectors means adding their entries

$$\begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{bmatrix} + \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix} = \begin{bmatrix} a_1 + b_1 \\ a_2 + b_2 \\ \vdots \\ a_n + b_n \end{bmatrix}$$

Vector Addition

adding two vectors means adding their entries

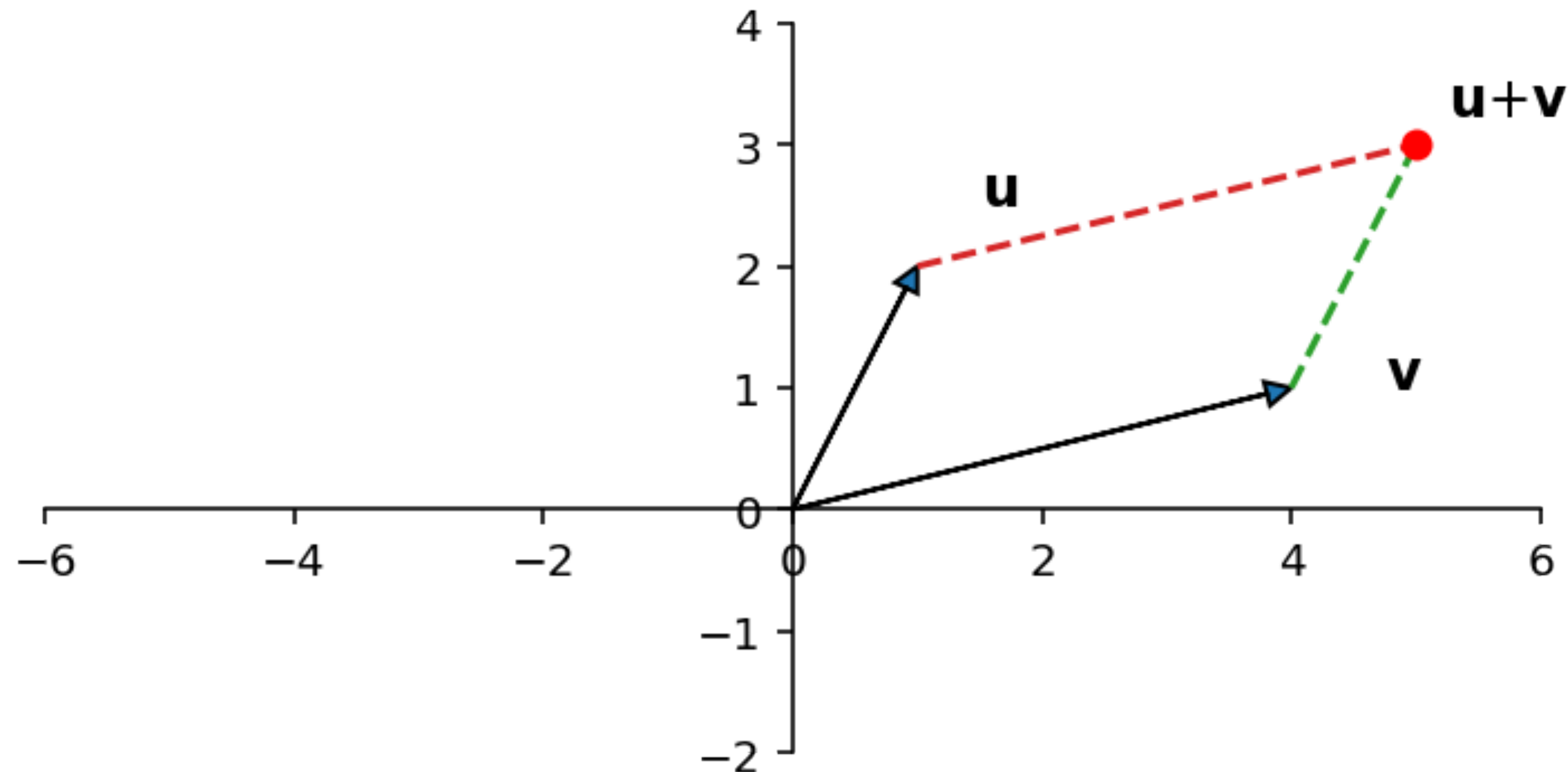
$$\begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{bmatrix} + \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix} = \begin{bmatrix} a_1 + b_1 \\ a_2 + b_2 \\ \vdots \\ a_n + b_n \end{bmatrix}$$

!! IMPORTANT !!

WE CAN ONLY ADD VECTORS OF THE SAME SIZE

Vector Addition (Geometrically)

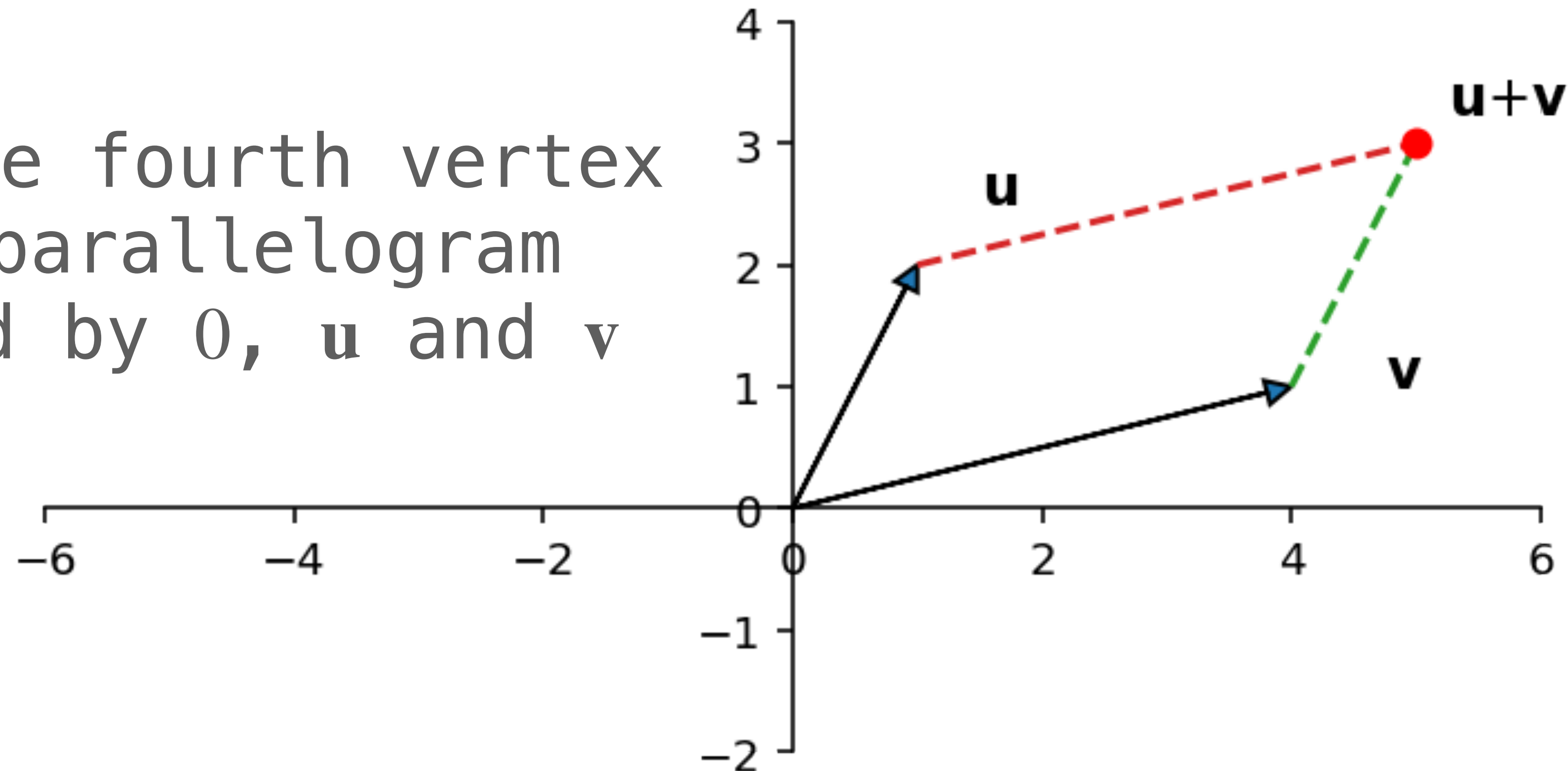
in $\mathbb{R}^2$ it's called the *parallelogram rule*



Vector Addition (Geometrically)

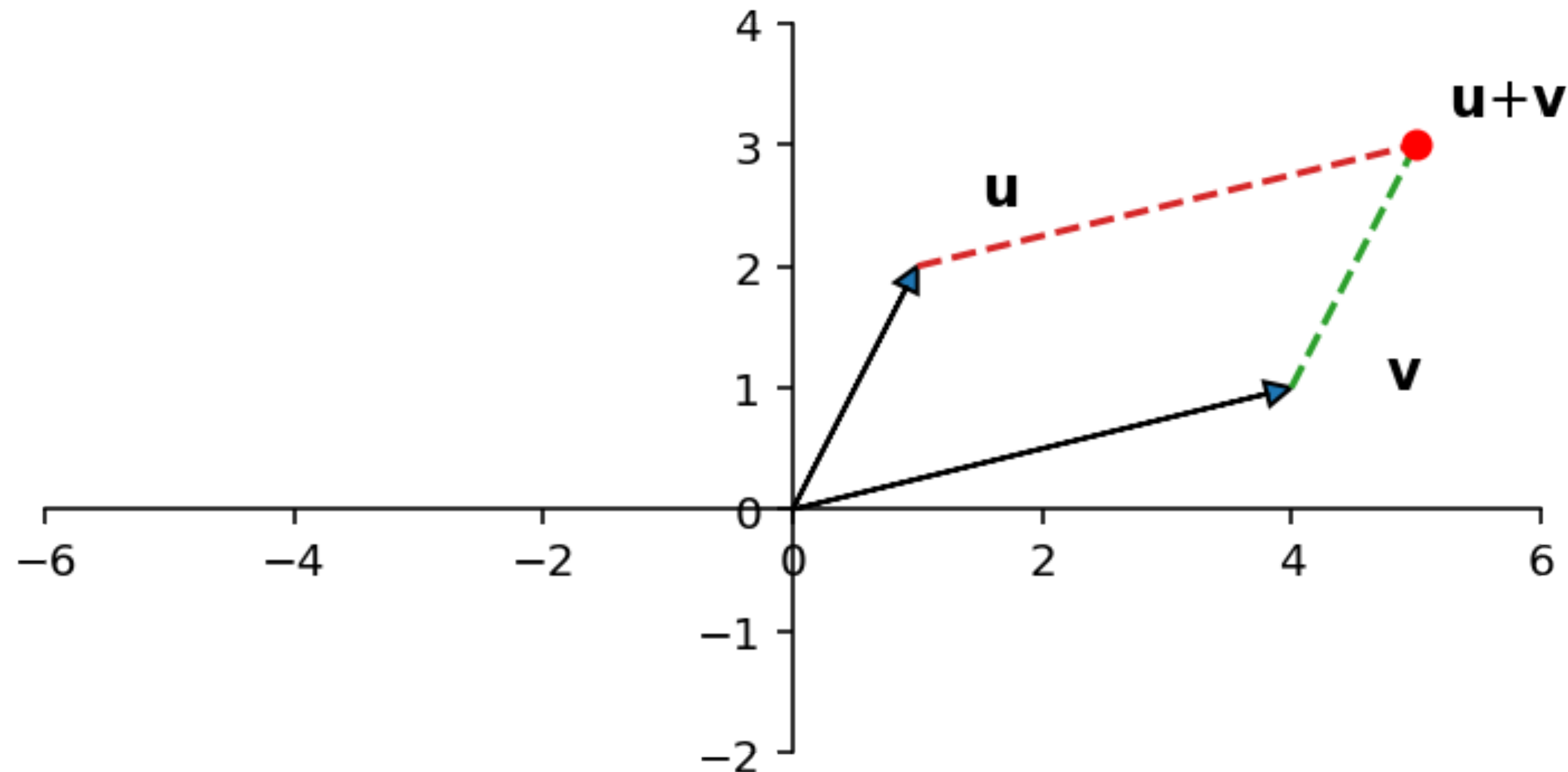
in $\mathbb{R}^2$ it's called the *parallelogram rule*

$\mathbf{u} + \mathbf{v}$ is the fourth vertex
of the parallelogram
generated by $\mathbf{0}$, $\mathbf{u}$ and $\mathbf{v}$



Vector Addition (Geometrically)

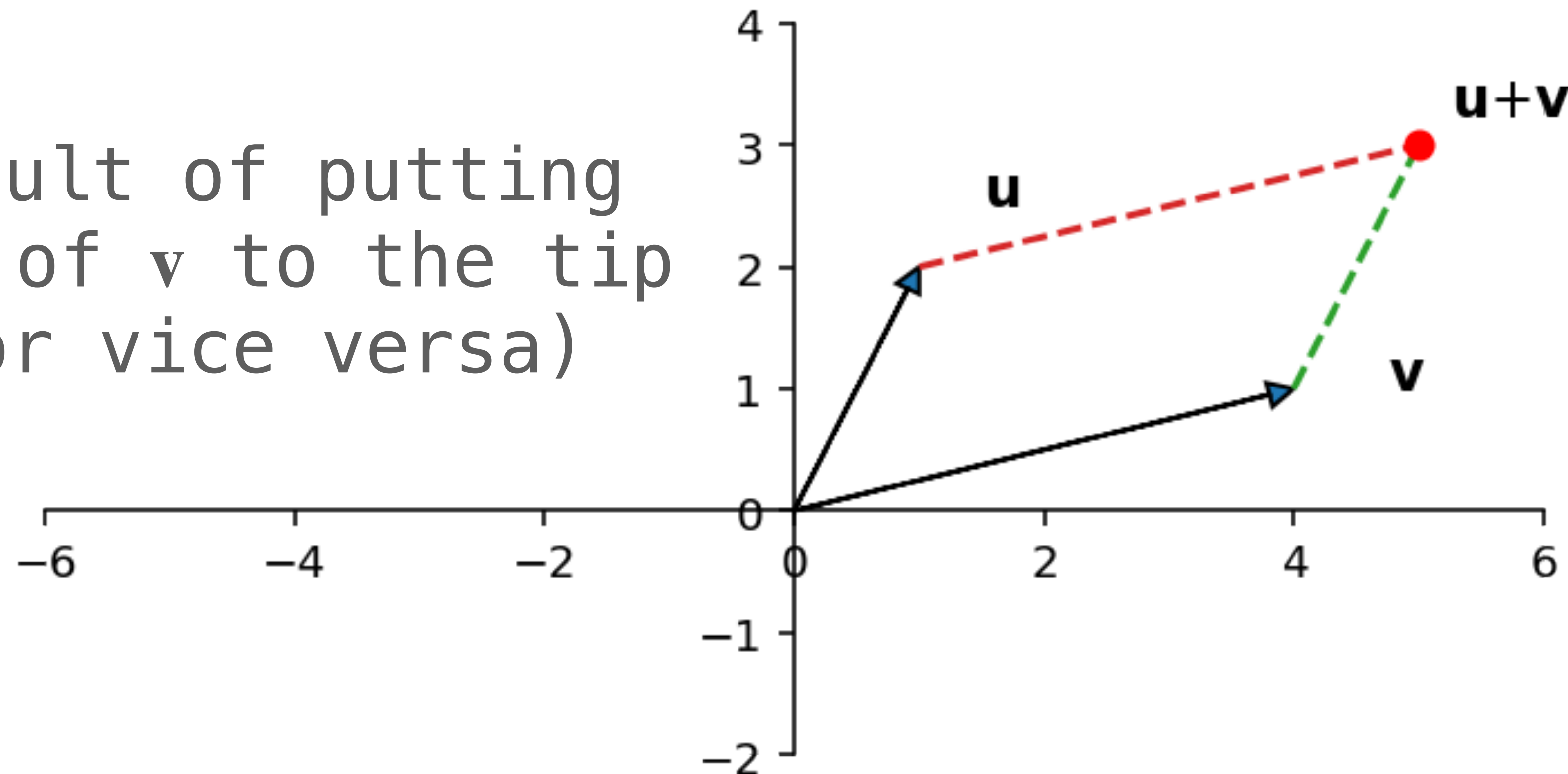
or the *tip-to-tail rule*



Vector Addition (Geometrically)

or the *tip-to-tail rule*

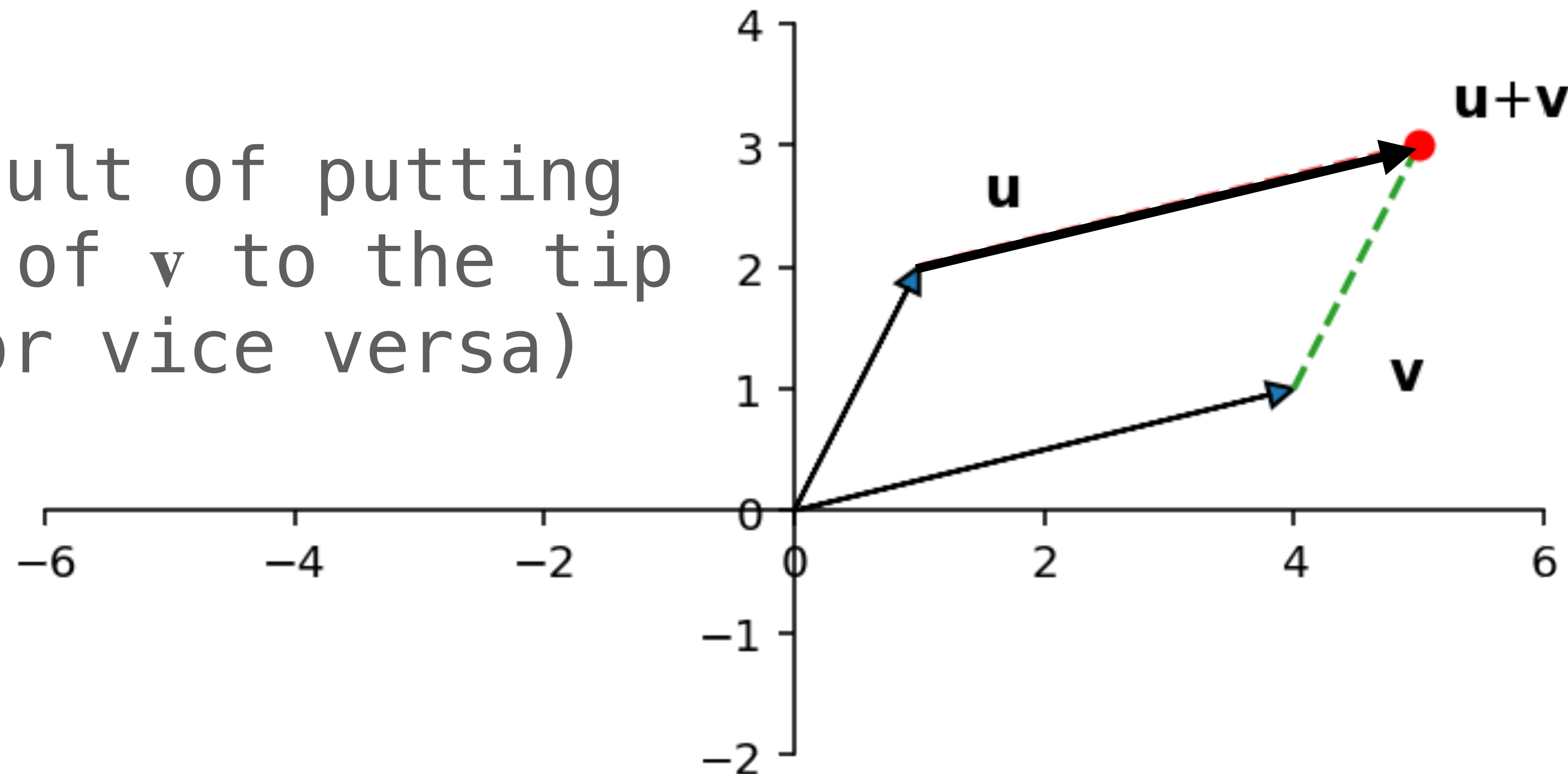
$\mathbf{u} + \mathbf{v}$ result of putting
the tail of $\mathbf{v}$ to the tip
of $\mathbf{u}$ (or vice versa)



Vector Addition (Geometrically)

or the *tip-to-tail rule*

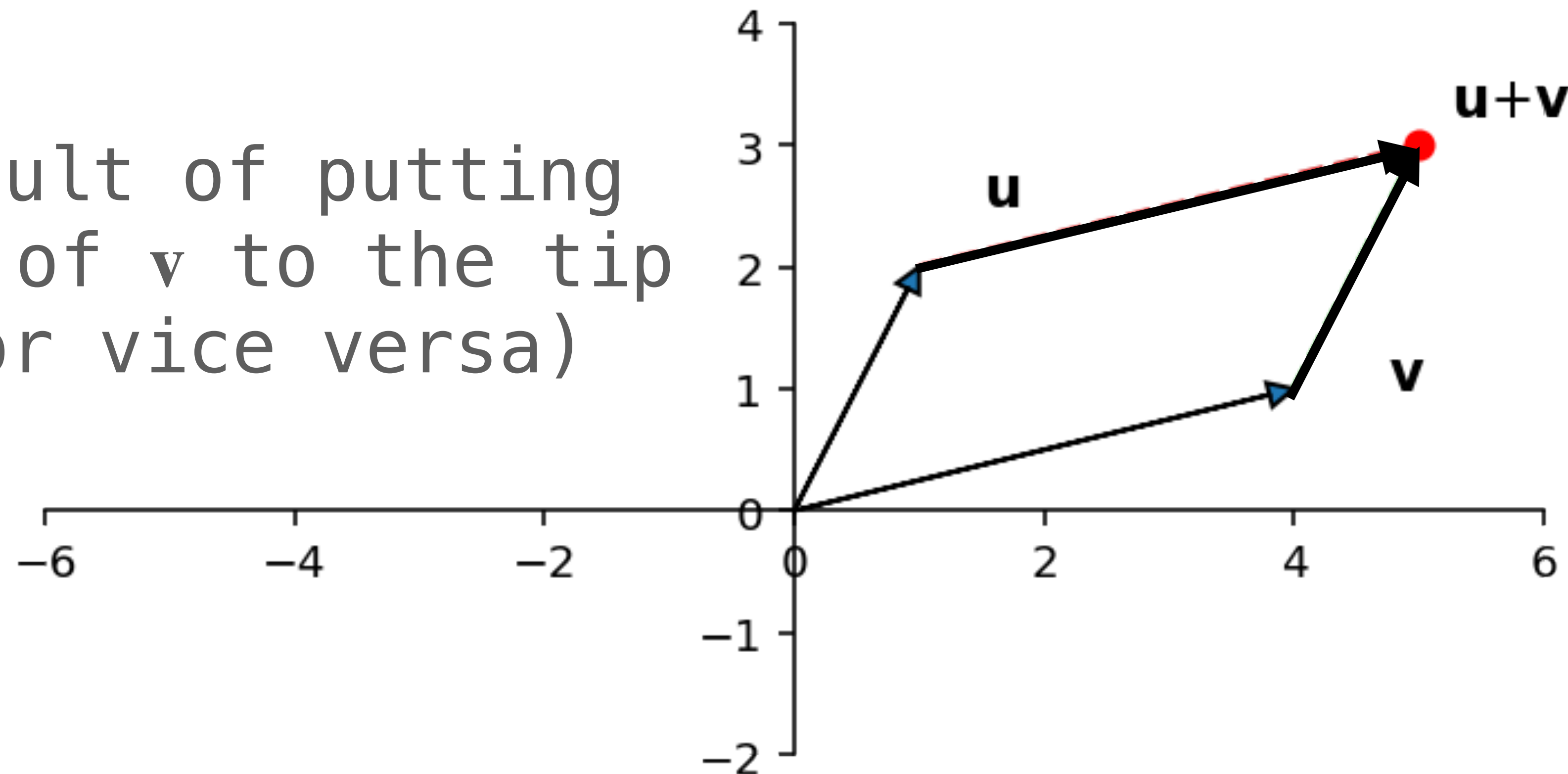
$\mathbf{u} + \mathbf{v}$ result of putting
the tail of $\mathbf{v}$ to the tip
of $\mathbf{u}$ (or vice versa)



Vector Addition (Geometrically)

or the *tip-to-tail rule*

$\mathbf{u} + \mathbf{v}$ result of putting
the tail of $\mathbf{v}$ to the tip
of $\mathbf{u}$ (or vice versa)



demo
(from ILA)

Vector "Interface"

equality what does it mean for two vectors to be equal?

addition what does $\mathbf{u} + \mathbf{v}$ (adding two vectors mean?

scaling what does $a\mathbf{v}$ (multiplying a vector by a real number) mean?

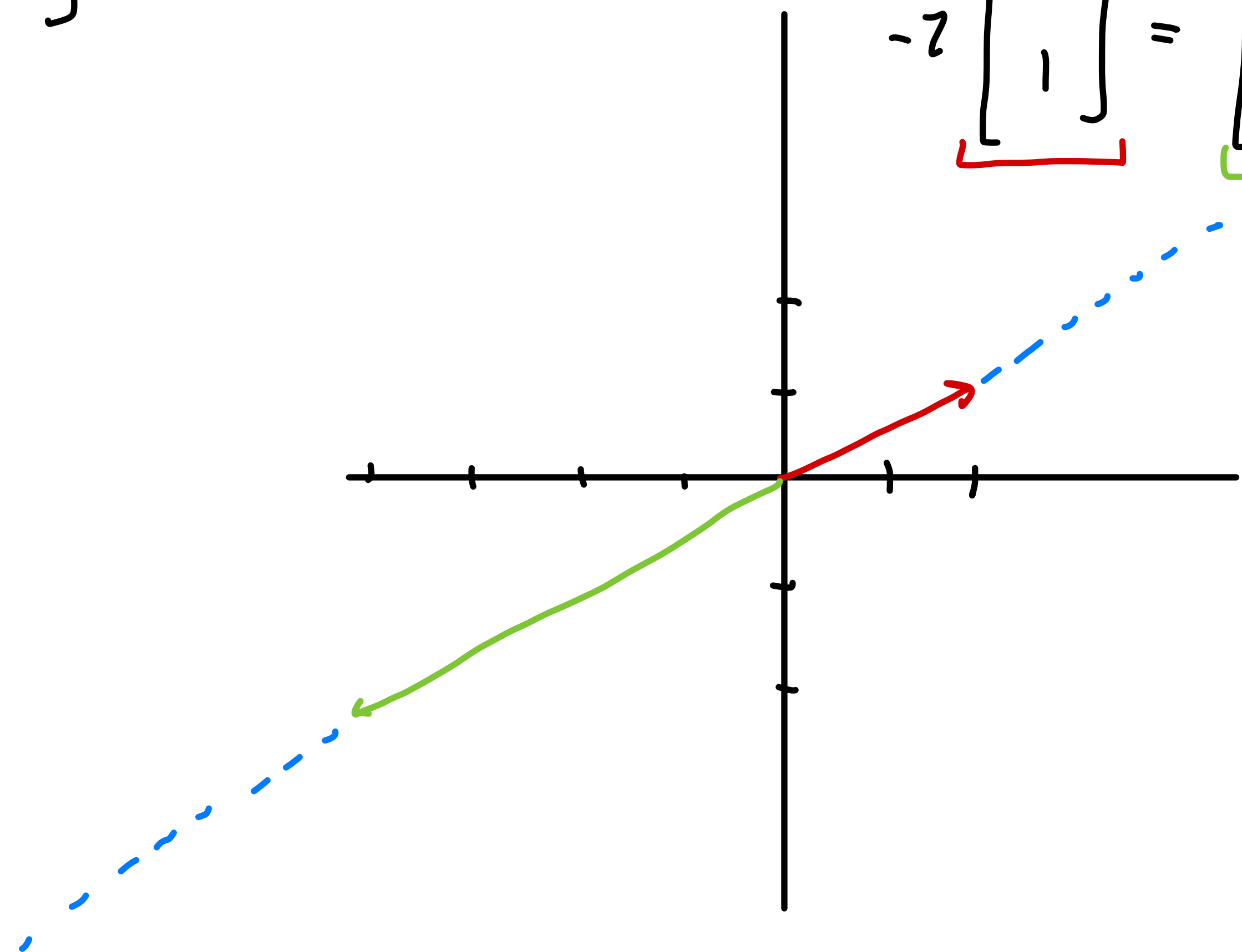
What properties do they need to satisfy?

Vector Scaling/Multiplication

$$2 \begin{bmatrix} 5 \\ 3 \end{bmatrix} = \begin{bmatrix} 2(5) \\ 2(3) \end{bmatrix} = \begin{bmatrix} 10 \\ 6 \end{bmatrix}$$

$$c \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} cx_1 \\ \vdots \\ cx_n \end{bmatrix}$$

$$-2 \begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} -4 \\ -2 \end{bmatrix}$$



Vector Scaling/Multiplication

scaling/multiplying a vector by a number means multiplying each of it's elements

$$a \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix} = \begin{bmatrix} ab_1 \\ ab_2 \\ \vdots \\ ab_n \end{bmatrix}$$

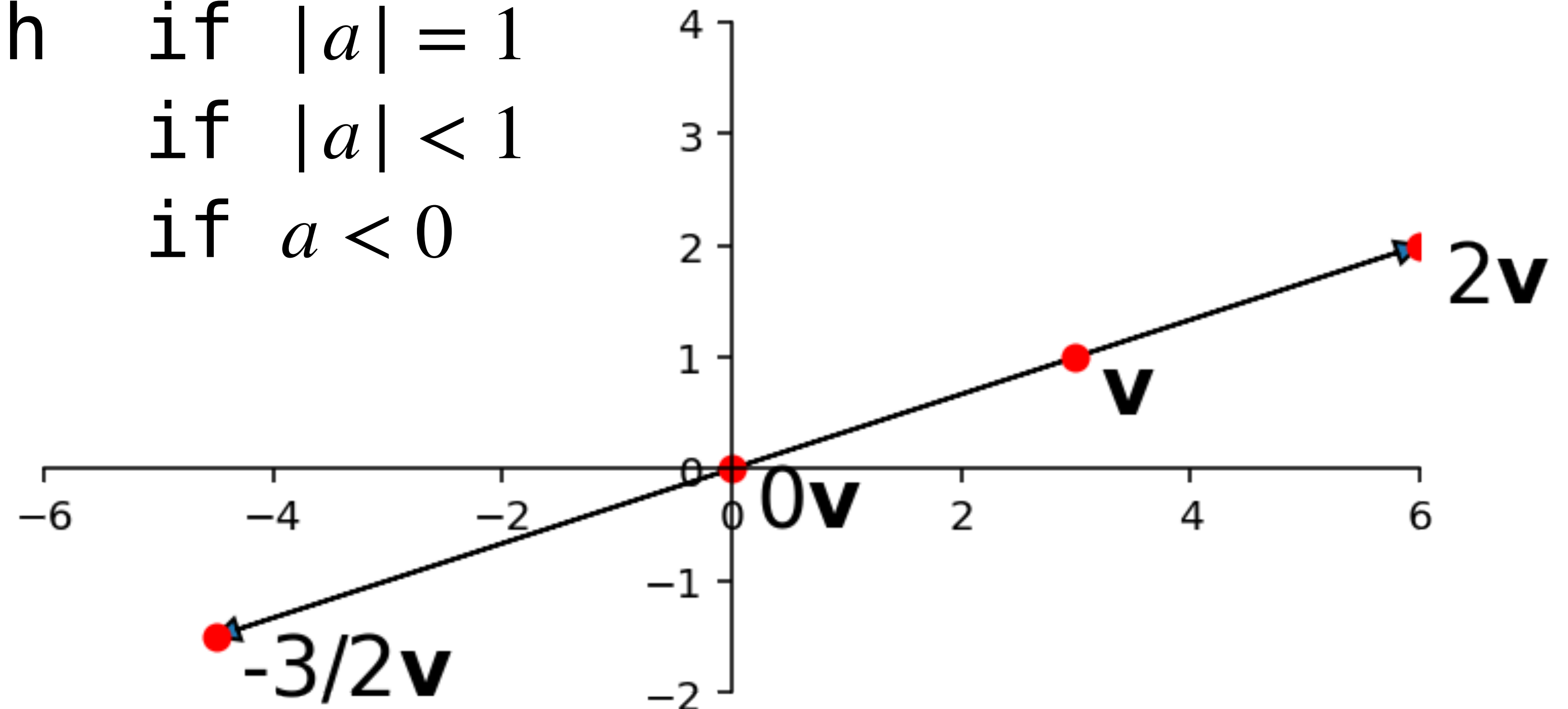
Vector Scaling/Multiplication (Example)

Scaling/multiplying a vector by a number means multiplying each of it's elements

$$3 \begin{bmatrix} 2 \\ 1 \\ 3.5 \\ 4 \end{bmatrix} = \begin{bmatrix} 3 \cdot 2 \\ 3 \cdot 1 \\ 3 \cdot 3.5 \\ 3 \cdot 4 \end{bmatrix} = \begin{bmatrix} 6 \\ 3 \\ 10.5 \\ 12 \end{bmatrix}$$

Vector Scaling (Geometrically)

longer	if $ a > 1$
the same length	if $ a = 1$
shorter	if $ a < 1$
reversed	if $a < 0$



demo
(from ILA)

Algebraic Properties

For any vectors $\mathbf{u}, \mathbf{v}, \mathbf{w}$ and any real numbers c, d :

$$\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$$

$$c(\mathbf{u} + \mathbf{v}) = c\mathbf{u} + c\mathbf{v}$$

$$(\mathbf{u} + \mathbf{v}) + \mathbf{w} = \mathbf{u} + (\mathbf{v} + \mathbf{w})$$

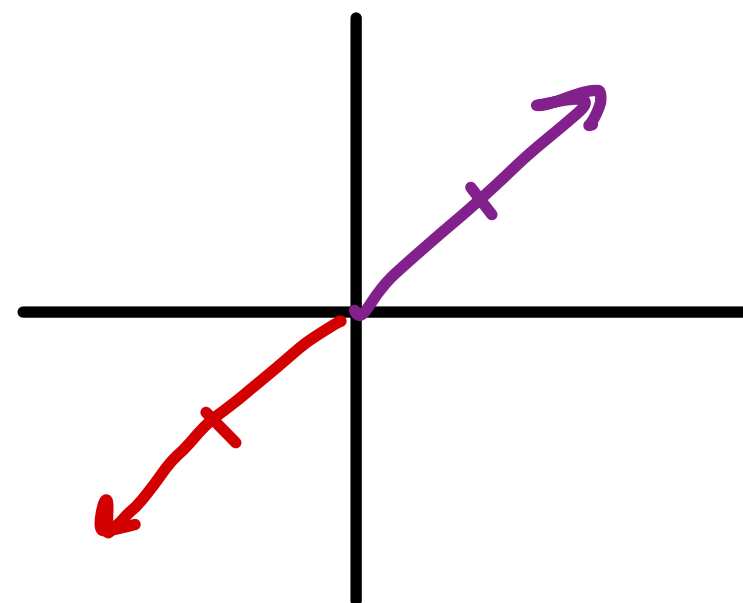
$$(c + d)\mathbf{u} = c\mathbf{u} + d\mathbf{u}$$

$$\mathbf{u} + \mathbf{0} = \mathbf{0} + \mathbf{u} = \mathbf{u}$$

$$c(d\mathbf{u}) = (cd)\mathbf{u}$$

$$\mathbf{u} + (-\mathbf{u}) = -\mathbf{u} + \mathbf{u} = \mathbf{0}$$

$$1\mathbf{u} = \mathbf{u}$$



Algebraic Properties

For any vectors $\mathbf{u}, \mathbf{v}, \mathbf{w}$ and any real numbers c, d :

$$\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$$

$$c(\mathbf{u} + \mathbf{v}) = c\mathbf{u} + c\mathbf{v}$$

$$(\mathbf{u} + \mathbf{v}) + \mathbf{w} = \mathbf{u} + (\mathbf{v} + \mathbf{w})$$

$$(c + d)\mathbf{u} = c\mathbf{u} + d\mathbf{u}$$

$$\mathbf{u} + \mathbf{0} = \mathbf{0} + \mathbf{u} = \mathbf{u}$$

$$c(d\mathbf{u}) = (cd)\mathbf{u}$$

$$\mathbf{u} + (-\mathbf{u}) = -\mathbf{u} + \mathbf{u} = \mathbf{0}$$

$$1\mathbf{u} = \mathbf{u}$$

these are requirements for any **vector space**
they matter more for *bizarre* vector spaces

$$\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$$

Example "Proof"

$$\begin{bmatrix} u_1 \\ \vdots \\ u_n \end{bmatrix} + \begin{bmatrix} v_1 \\ \vdots \\ v_n \end{bmatrix} = \begin{bmatrix} u_1 + v_1 \\ \vdots \\ u_n + v_n \end{bmatrix} = \begin{bmatrix} v_1 + u_1 \\ \vdots \\ v_n + u_n \end{bmatrix} = \begin{bmatrix} v_1 \\ \vdots \\ v_n \end{bmatrix} + \begin{bmatrix} u_1 \\ \vdots \\ u_n \end{bmatrix}$$

Question (Practice)

$$3 \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} + 2 \begin{bmatrix} 2 \\ 0 \\ 3 \\ -1 \end{bmatrix} - \begin{bmatrix} -3 \\ 4 \\ 2 \\ 0 \end{bmatrix}$$

Compute the value of the above vector

Answer

$$3 \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} + 2 \begin{bmatrix} 2 \\ 0 \\ 3 \\ -1 \end{bmatrix} - \begin{bmatrix} -3 \\ 4 \\ 2 \\ 0 \end{bmatrix}$$

In Sum

In Sum

we can add vectors

In Sum

we can add vectors

we can scale vectors

In Sum

we can add vectors

we can scale vectors

What vectors can we make with these operations?

Linear Combinations

Linear Combinations

Df. A lin. comb. of $v_1, \dots, v_n$ is a weighted
sum of $v_1, \dots, v_n$

$$c_1 \vec{v}_1 + \dots + c_n v_n = \sum_{i=1}^n c_i \vec{v}_i$$

weights

$$\begin{bmatrix} 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 5 \\ 3 \end{bmatrix}$$

ex.

$$2 \begin{bmatrix} 1 \\ 1 \end{bmatrix} + 3 \begin{bmatrix} 5 \\ 3 \end{bmatrix} = \begin{bmatrix} 2+15 \\ 2+9 \end{bmatrix} = \begin{bmatrix} 17 \\ 11 \end{bmatrix}$$

is a lin comb. of $\begin{bmatrix} 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 5 \\ 3 \end{bmatrix}$

Linear Combinations

Df. a *linear combination* of vectors

$$\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$$

is a vector of the form

$$\alpha_1 \mathbf{v}_1 + \alpha_2 \mathbf{v}_2 + \dots + \alpha_n \mathbf{v}_n$$

where $\alpha_1, \alpha_2, \dots, \alpha_n$ are in $\mathbb{R}$

Linear Combinations

Df. a *linear combination* of vectors

$$\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$$

is a vector of the form

$$\alpha_1 \mathbf{v}_1 + \alpha_2 \mathbf{v}_2 + \dots + \alpha_n \mathbf{v}_n$$

where $\alpha_1, \alpha_2, \dots, \alpha_n$ are in $\mathbb{R}$
weights

Linear Combinations

Df. a *linear combination* of vectors

$$\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$$

is a vector of the form

$$\alpha_1 \mathbf{v}_1 + \alpha_2 \mathbf{v}_2 + \dots + \alpha_n \mathbf{v}_n$$

Looks suspiciously like
a linear equation...

where $\alpha_1, \alpha_2, \dots, \alpha_n$ are in $\mathbb{R}$
weights

Linear Combinations (Example)

$$3 \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} + 2 \begin{bmatrix} 2 \\ 0 \\ 3 \\ -1 \end{bmatrix} - \begin{bmatrix} -3 \\ 4 \\ 2 \\ 0 \end{bmatrix}$$

Definition. a *linear combination* of vectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$ is a vector of the form

$$\alpha_1 \mathbf{v}_1 + \alpha_2 \mathbf{v}_2 + \dots + \alpha_n \mathbf{v}_n$$

where $\alpha_1, \alpha_2, \dots, \alpha_n$ are in $\mathbb{R}$

demo
(from ILA)

The Fundamental Concern

$$2 \begin{bmatrix} 1 \\ 2 \end{bmatrix} - 3 \begin{bmatrix} -1 \\ -1 \end{bmatrix} = \begin{bmatrix} 5 \\ 7 \end{bmatrix}$$

Can $\mathbf{u}$ be written as a linear combination of $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$?

That is, are there weights $\alpha_1, \alpha_2, \dots, \alpha_n$ such that $\alpha_1 \mathbf{v}_1 + \alpha_2 \mathbf{v}_2 + \dots \alpha_n \mathbf{v}_n = \mathbf{u}$?

Why is this fundamental?

I'm going to ask that you suspend your disbelief...

Why is this fundamental?

I'm going to ask that you suspend your disbelief...

For now, how do we solve this problem?

Vector Equations and Linear Systems

The Fundamental Connection

$$x_1 \begin{bmatrix} 1 \\ -2 \\ -5 \end{bmatrix} + x_2 \begin{bmatrix} 2 \\ 5 \\ 6 \end{bmatrix} = \begin{bmatrix} 7 \\ 4 \\ -3 \end{bmatrix}$$

The Fundamental Connection

$$x_1 \begin{bmatrix} 1 \\ -2 \\ -5 \end{bmatrix} + x_2 \begin{bmatrix} 2 \\ 5 \\ 6 \end{bmatrix} = \begin{bmatrix} 7 \\ 4 \\ -3 \end{bmatrix}$$

We want the weights...

The Fundamental Connection

$$x_1 \begin{bmatrix} 1 \\ -2 \\ -5 \end{bmatrix} + x_2 \begin{bmatrix} 2 \\ 5 \\ 6 \end{bmatrix} = \begin{bmatrix} 7 \\ 4 \\ -3 \end{bmatrix}$$

We want the weights...

What if we write them as unknowns?

Some Symbol Pushing...

$$x_1 \begin{bmatrix} 1 \\ -2 \\ -5 \end{bmatrix} + x_2 \begin{bmatrix} 2 \\ 5 \\ 6 \end{bmatrix} = \begin{bmatrix} 7 \\ 4 \\ -3 \end{bmatrix}$$

scaling

$$\begin{bmatrix} x_1 \\ -2x_1 \\ -5x_1 \end{bmatrix} + \begin{bmatrix} 2x_2 \\ 5x_2 \\ 6x_2 \end{bmatrix} = \begin{bmatrix} 7 \\ 4 \\ -3 \end{bmatrix}$$

addition

$$\begin{bmatrix} x_1 + 2x_2 \\ -2x_1 + 5x_2 \\ -5x_1 + 6x_2 \end{bmatrix} = \begin{bmatrix} 7 \\ 4 \\ -3 \end{bmatrix}$$

equality

$$\begin{aligned} x_1 + 2x_2 &= 7 \\ -2x_1 + 5x_2 &= 4 \\ -5x_1 + 6x_2 &= -3 \end{aligned}$$

linear system

Some Symbol Pushing...

$$x_1 \begin{bmatrix} 1 \\ -2 \\ -5 \end{bmatrix} + x_2 \begin{bmatrix} 2 \\ 5 \\ 6 \end{bmatrix} = \begin{bmatrix} 7 \\ 4 \\ -3 \end{bmatrix}$$

Some Symbol Pushing...

$$\begin{bmatrix} x_1 \\ (-2)x_1 \\ (-5)x_1 \end{bmatrix} + \begin{bmatrix} 2x_2 \\ 5x_2 \\ 6x_2 \end{bmatrix} = \begin{bmatrix} 7 \\ 4 \\ -3 \end{bmatrix}$$

Some Symbol Pushing...

$$\begin{bmatrix} x_1 + 2x_2 \\ (-2)x_1 + 5x_2 \\ -5x_1 + 6x_2 \end{bmatrix} = \begin{bmatrix} 7 \\ 4 \\ -3 \end{bmatrix}$$

Some Symbol Pushing...

$$x_1 + 2x_2 = 7$$

$$(-2)x_1 + 5x_2 = 4$$

$$-5x_1 + 6x_2 = -3$$

Some Symbol Pushing...

$$x_1 + 2x_2 = 7$$

$$(-2)x_1 + 5x_2 = 4$$

$$-5x_1 + 6x_2 = -3$$

we get a system
of linear
equations we
know how to
solve

General Vector Equations

$$x_1 \begin{bmatrix} a_{11} \\ a_{21} \\ \vdots \\ a_{1m} \end{bmatrix} + x_2 \begin{bmatrix} a_{21} \\ a_{21} \\ \vdots \\ a_{2m} \end{bmatrix} + \dots + x_n \begin{bmatrix} a_{n1} \\ a_{n2} \\ \vdots \\ a_{nm} \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix}$$

General Vector Equations

$$\begin{bmatrix} a_{11}x_1 \\ a_{21}x_1 \\ \vdots \\ a_{1m}x_1 \end{bmatrix} + \begin{bmatrix} a_{21}x_2 \\ a_{21}x_2 \\ \vdots \\ a_{2m}x_1 \end{bmatrix} + \dots + \begin{bmatrix} a_{n1}x_n \\ a_{n2}x_n \\ \vdots \\ a_{nm}x_n \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix}$$

by vector scaling

General Vector Equations

$$\begin{bmatrix} a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n \\ a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n \\ \vdots \\ a_{m1}x_1 + a_{m2}x_2 + \dots + a_{mn}x_n \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix}$$

by vector addition

General Vector Equations

$$a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n = b_1$$

$$a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n = b_2$$

$$\vdots$$

$$a_{m1}x_1 + a_{m2}x_2 + \dots + a_{mn}x_n = b_m$$

by vector equality

The Fundamental Connection

$$\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} & b_1 \\ a_{21} & a_{22} & \cdots & a_{2n} & b_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} & b_m \end{bmatrix}$$

augmented matrix

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n &= b_1 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n &= b_2 \\ &\vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n &= b_m \end{aligned}$$

system of linear equations

$$x_1 \begin{bmatrix} a_{11} \\ a_{21} \\ \vdots \\ a_{m1} \end{bmatrix} + x_2 \begin{bmatrix} a_{12} \\ a_{22} \\ \vdots \\ a_{m2} \end{bmatrix} + \cdots + x_n \begin{bmatrix} a_{1n} \\ a_{2n} \\ \vdots \\ a_{mn} \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix}$$

vector equation

The Fundamental Connection

$$\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} & b_1 \\ a_{21} & a_{22} & \cdots & a_{2n} & b_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} & b_m \end{bmatrix}$$

augmented matrix

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n &= b_1 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n &= b_2 \\ &\vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n &= b_m \end{aligned}$$

system of linear equations

$$x_1 \begin{bmatrix} a_{11} \\ a_{21} \\ \vdots \\ a_{m1} \end{bmatrix} + x_2 \begin{bmatrix} a_{12} \\ a_{22} \\ \vdots \\ a_{m2} \end{bmatrix} + \cdots + x_n \begin{bmatrix} a_{1n} \\ a_{2n} \\ \vdots \\ a_{mn} \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix}$$

vector equation

The Fundamental Connection

$$\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} & b_1 \\ a_{21} & a_{22} & \cdots & a_{2n} & b_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} & b_m \end{bmatrix}$$

augmented matrix

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n &= b_1 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n &= b_2 \\ &\vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n &= b_m \end{aligned}$$

system of linear equations

$$x_1 \begin{bmatrix} a_{11} \\ a_{21} \\ \vdots \\ a_{m1} \end{bmatrix} + x_2 \begin{bmatrix} a_{12} \\ a_{22} \\ \vdots \\ a_{m2} \end{bmatrix} + \cdots + x_n \begin{bmatrix} a_{1n} \\ a_{2n} \\ \vdots \\ a_{mn} \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix}$$

vector equation

The Fundamental Connection

$$\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} & b_1 \\ a_{21} & a_{22} & \cdots & a_{2n} & b_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} & b_m \end{bmatrix}$$

augmented matrix

*This is our big
perspective shift
of the day*

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n &= b_1 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n &= b_2 \\ &\vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n &= b_m \end{aligned}$$

system of linear equations

$$x_1 \begin{bmatrix} a_{11} \\ a_{21} \\ \vdots \\ a_{m1} \end{bmatrix} + x_2 \begin{bmatrix} a_{12} \\ a_{22} \\ \vdots \\ a_{m2} \end{bmatrix} + \cdots + x_n \begin{bmatrix} a_{1n} \\ a_{2n} \\ \vdots \\ a_{mn} \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix}$$

vector equation

How To: Linear Combination Problems

How To: Linear Combination Problems

Q. Can $\mathbf{b}$ be written as a linear combination of $\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_n$?

How To: Linear Combination Problems

Q. Can $\mathbf{b}$ be written as a linear combination of $\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_n$?

A. Solve the system of linear equations with the augmented matrix

$$[\mathbf{a}_1 \quad \mathbf{a}_2 \quad \dots \quad \mathbf{a}_n \quad \mathbf{b}]$$

How To: Linear Combination Problems

Q. Can $\mathbf{b}$ be written as a linear combination of $\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_n$?

A. Solve the system of linear equations with the augmented matrix

$$[\mathbf{a}_1 \quad \mathbf{a}_2 \quad \dots \quad \mathbf{a}_n \quad \mathbf{b}]$$

A solution to this system is a set of weights to define $\mathbf{b}$ as a linear combination of $\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_n$

How To: Linear Combination Problems

Q. Can $\mathbf{b}$ be written as a linear combination of $\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_n$?

A. Solve the system of linear equations with the augmented matrix

$$\left[\begin{array}{c|c|c|c|c} | & | & & | & | \\ \mathbf{a}_1 & \mathbf{a}_2 & \dots & \mathbf{a}_n & \mathbf{b} \\ | & | & & | & | \end{array} \right]$$

← this is notation for
building a matrix
out of column
vectors

A solution to this system is a set of weights to define $\mathbf{b}$ as a linear combination of $\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_n$

Question

Can $\begin{bmatrix} 7 \\ 4 \\ -3 \end{bmatrix}$ be written as a linear combination of $\begin{bmatrix} 1 \\ -2 \\ -5 \end{bmatrix}$ and $\begin{bmatrix} 2 \\ 5 \\ 6 \end{bmatrix}$?

$$x_1 \begin{bmatrix} 1 \\ -2 \\ -5 \end{bmatrix} + x_2 \begin{bmatrix} 2 \\ 5 \\ 6 \end{bmatrix} = \begin{bmatrix} 7 \\ 4 \\ -3 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 & 7 \\ -2 & 5 & 4 \\ -5 & 6 & -3 \end{bmatrix} \sim \text{RREF}$$

exercise.

Answer

Spans

Spans

Spans

Spans

Df. the *span* of a set of vectors is the set of all possible linear combinations of them

Spans

Df. the *span* of a set of vectors is the set of all possible linear combinations of them

$$\text{span}\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\} = \{\alpha_1\mathbf{v}_1 + \alpha_2\mathbf{v}_2 + \dots\alpha_n\mathbf{v}_n : \alpha_1, \alpha_2, \dots, \alpha_n \text{ are in } \mathbb{R}\}$$

Spans

Df. the *span* of a set of vectors is the set of all possible linear combinations of them

$$\text{span}\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\} = \{\alpha_1\mathbf{v}_1 + \alpha_2\mathbf{v}_2 + \dots\alpha_n\mathbf{v}_n : \alpha_1, \alpha_2, \dots, \alpha_n \text{ are in } \mathbb{R}\}$$

$\mathbf{u} \in \text{span}\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ exactly when $\mathbf{u}$ can be expressed as a linear combination of those vectors

Spans

Df. the *span* of a set of vectors is the set of all possible linear combinations of them

$$\text{span}\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\} = \{\alpha_1 \mathbf{v}_1 + \alpha_2 \mathbf{v}_2 + \dots \alpha_n \mathbf{v}_n : \alpha_1, \alpha_2, \dots, \alpha_n \text{ are in } \mathbb{R}\}$$

read: $\mathbf{u}$ is an element of $\text{span}\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$



$\mathbf{u} \in \text{span}\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ exactly when $\mathbf{u}$ can be expressed as a linear combination of those vectors

$$\text{span} \left\{ \begin{bmatrix} 3 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} -1 \\ 0 \\ 0 \end{bmatrix} \right\}$$

$$\begin{bmatrix} 3 \\ 1 \\ 2 \end{bmatrix} \notin \text{span} \left\{ \begin{bmatrix} 3 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} -1 \\ 0 \\ 0 \end{bmatrix} \right\}$$

$$\begin{bmatrix} 2 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 \\ 1 \\ 1 \end{bmatrix} + \begin{bmatrix} -1 \\ 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 2 \\ 1 \\ 1 \end{bmatrix} \in \text{span} \left\{ \begin{bmatrix} 3 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} -1 \\ 0 \\ 0 \end{bmatrix} \right\}$$

$$\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 \\ 1 \\ 1 \end{bmatrix} - 2 \begin{bmatrix} -1 \\ 0 \\ 0 \end{bmatrix}$$

$$x_1 \begin{bmatrix} 3 \\ 1 \\ 1 \end{bmatrix} + x_2 \begin{bmatrix} -1 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 3x_1 - x_2 \\ x_1 \\ x_1 \end{bmatrix}$$

demo
(from ILA)

Span of One Vector

Span of Two Vector

Spans (Geometrically)

Spans (Geometrically)

for one vector

Spans (Geometrically)

for one vector

$$\text{span}\{\mathbf{v}\} = \{\alpha\mathbf{v} : \alpha \in \mathbb{R}\}$$

Spans (Geometrically)

for one vector

$$\text{span}\{\mathbf{v}\} = \{\alpha\mathbf{v} : \alpha \in \mathbb{R}\}$$

this is all scalar multiple of $\mathbf{v}$

Spans (Geometrically)

for one vector

$$\text{span}\{\mathbf{v}\} = \{\alpha\mathbf{v} : \alpha \in \mathbb{R}\}$$

this is **all scalar multiple of $\mathbf{v}$**

the span of one vector is a **line**

Spans (Geometrically)

Spans (Geometrically)

the span of **two** vectors can be a **plane**

Spans (Geometrically)

the span of **two** vectors can be a **plane**

the span of **three** vectors can be a **hyperplane**

Spans (Geometrically)

the span of **two** vectors can be a **plane**

the span of **three** vectors can be a **hyperplane**

!!IMPORTANT!!

In all cases they pass through the origin

demo
(from ILA)

How To: Span Problems

How To: Span Problems

Question. Is $\mathbf{b} \in \text{span}\{\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_n\}$?

How To: Span Problems

Question. Is $\mathbf{b} \in \text{span}\{\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_n\}$?

Solution. Determine if $\mathbf{b}$ can be written as a linear combination of $\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_n$

How To: Span Problems

Question. Is $\mathbf{b} \in \text{span}\{\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_n\}$?

Solution. Determine if $\mathbf{b}$ can be written as a linear combination of $\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_n$

you know how to do this now

Example

$$\text{Is } \begin{bmatrix} 7 \\ 4 \\ -3 \end{bmatrix} \text{ in span } \left\{ \begin{bmatrix} 1 \\ -2 \\ -5 \end{bmatrix}, \begin{bmatrix} 2 \\ 5 \\ 6 \end{bmatrix} \right\} ?$$

Question (Conceptual)

What does it mean geometrically if $\mathbf{b} \notin \text{span}\{\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_n\}$?

demo
(from ILA)

How To: Inconsistency and Spans

How To: Inconsistency and Spans

Q. find a vector $\mathbf{b}$ which *does not* appear in
 $\text{span}\{\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_n\}$

How To: Inconsistency and Spans

Q. find a vector $\mathbf{b}$ which *does not* appear in $\text{span}\{\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_n\}$

A. Choose $\mathbf{b}$ so that $[\mathbf{a}_1 \ \mathbf{a}_2 \ \dots \ \mathbf{a}_n \ \mathbf{b}]$ is the augmented matrix of an *inconsistent* system

How To: Inconsistency and Spans

Q. find a vector $\mathbf{b}$ which *does not* appear in $\text{span}\{\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_n\}$

A. Choose $\mathbf{b}$ so that $[\mathbf{a}_1 \ \mathbf{a}_2 \ \dots \ \mathbf{a}_n \ \mathbf{b}]$ is the augmented matrix of an *inconsistent* system

There is **no way** to write $\mathbf{b}$ as a linear combination

Example

*Find a vector **not** in* $\text{span} \left\{ \begin{bmatrix} 2 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 5 \\ 3 \\ 3 \end{bmatrix} \right\}$

Summary

Vectors are fundamental objects

We can think of them as the **columns** of a linear system

We can **scale** them and **add** them together

They can **span** spaces which represent **hyperplanes**